AI Use Declaration Form

Course: Introduction to Artificial Intelligence
Lab Title: Lab 1 Part A — Uninformed Search Algorithms
Student Name: _Zahara Seybou Harouna____
Student ID: _30862027____
GitHub Repository Link:
https://github.com/zaharaSHarouna/lab-2-predictive-analytics.git____ ___
Date Submitted: _03/07/2026____

1. AI Use Summary

Question	**Student Response**
Did you use any AI tool for this lab?	**Yes**
Estimated percentage of the work influenced by AI	19%
Did you attach evidence of AI use where applicable?	**Yes**

2. Details of AI Use

Complete the table below for each AI tool used. Add more rows where necessary.

Name of AI Tool Used	**Purpose for Using the Tool**	**Prompt or Instruction Given to the Tool**	**Part(s) of the Work Influenced by the Tool**	**How I Verified, Edited, or Improved the AI Output**
ChatGpt	I have a little knowledge about Obesity specially when it come to clinical setting so I acquired to understand , that helped me to do understand the second part of the lab.	How can you explain Obesity for me if i want to do insupervided learing on it in clinical setting	Second part	I just learned.

Claude ai	When I reached the Elbow part in clustering the result of K I got put me in doubt, So asked if my code wasn't correct but my code was actually correct, from the Elbow method I got K=2 that didn't make sense for the overall data.	K-Means fitted with K=2 Cluster sizes: Cluster 0: 702 points Cluster 1: 1409 points. I shared my data and the code of that part.	Second part	I didn't my code. was correct!

3. Attachment of AI Output Evidence

Where required, attach evidence of AI use. Tick all that apply. Add more rows where necessary.

Evidence Type	**Attached? (Yes, No, N/A)**	**File Name**
AI-generated draft	yes	In AI use Declaration from below
Prompt history	yes	In AI use Declaration from below
Screenshot of AI interaction	yes	In AI use Declaration from below
Exported AI conversation	Yes	In AI use Declaration from below
AI-generated code snippet	-	-
Revised version showing student input	-	-

4. Student Declaration I declare that:

Declaration Statement	**Response (Yes/No)**

The submitted work is my own work.	
Any use of AI tools has been clearly declared in this form.	yes
The prompts, instructions, and AI-generated outputs have been disclosed where applicable.	yes
I reviewed, tested, edited, and improved any AI-generated content before submission.	Yes,
My AI usage does not exceed 25% of the entire work.	Yes
I understand that undeclared or excessive AI use may be treated as academic misconduct.	Yes

AI prompts: See the table above.

```
K-Means fitted with K=2
Cluster sizes:
  Cluster 0: 702 points
  Cluster 1: 1409 points
```

Good thinking! 😰 But let's weigh the pros and cons before deciding:

K=2 vs K=7 — Which Should We Use?

	K=2	K=7
Silhouette score	✅ Best (0.1879)	⚠️ Lower
Matches true	❌ Only 2 groups vs 7 real ones	✅ Matches real obesity levels

K=2 vs K=7 — Which Should We Use?

	K=2	K=7
Silhouette score	✅ Best (0.1879)	⚠️ Lower
Matches true classes	❌ Only 2 groups vs 7 real ones	✅ Matches real obesity levels
Crosstab comparison	❌ Meaningless (can't compare 2 clusters to 7 classes)	✅ Meaningful comparison
Lab requirement	❌ Lab asks to compare clusters to true obesity levels	✅ Fulfills lab requirement
Clinical value	⚠️ Only "at risk" vs "not at ri	✅ 7 actionable intervention groups

How can you explain Obesity for me if i want to do insupervided learing on it in clinical setting

If you want to explain **obesity** for **unsupervised learning in a clinical setting** (for example, using machine learning to discover patterns in patient data without predefined labels), you can present it like this:

Obesity: A Clinical Overview

Obesity is a **chronic, complex disease** characterized by excessive accumulation of body fat that increases the risk of adverse health outcomes. It is commonly assessed using **Body Mass Index (BMI)**, although other measures such as waist circumference and body fat percentage provide additional information.

BMI Classification (Adults):

- Underweight: <18.5 kg/m²
- Normal weight: 18.5–24.9 kg/m²
- Overweight: 25.0–29.9 kg/m²
- Obesity Class I: 30.0–34.9 kg/m²

Obesity in Unsupervised Learning

In a clinical machine learning context, **unsupervised learning** is used to identify hidden patterns in obesity-related patient data **without using predefined outcome labels**.

Example Clinical Dataset

Variables may include:

- Age
- Sex
- BMI
- Waist circumference
- Blood pressure
- Fasting blood glucose
- Cholesterol levels
- Physical activity level
- Daily calorie intake
- Sleep duration

Normaly, how many Obesity group could we have

The number of **obesity groups** depends on what you mean:

1. Clinical classification (most commonly used)

Clinically, obesity is usually divided into **3 groups (classes)** based on **Body Mass Index (BMI)**:

Group	BMI (kg/m²)
Obesity Class I	30.0–34.9
Obesity Class II	35.0–39.9
Obesity Class III (severe obesity)	≥40.0

Student Signature: __ZaharaSeybou____

Date: ____03/07/2026____